

MINISTRY OF EDUCATION AND TRAINING

FPT UNIVERSITY

DEPARTMENT OF ITS



FPT UNIVERSITY

GRADUATE THESIS REPORT

**ANOMALOUS HUMAN ACTIVITY DETECTION
BY WEAKLY SUPERVISED APPROACH**

MAJOR: ARTIFICIAL INTELLIGENCE
CAPSTONE CODE: SP25AI39

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Abstract

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A Introduction

With the increasing prevalence of technology in everyday life, the integration of AI-powered systems has become indispensable across various domains. One such application is Video Anomaly Detection (VAD)—a field aimed at automating the identification of unusual or suspicious activities within video footage [2]. Despite the progress achieved in this area, challenges such as the exhaustive effort required for manual annotation of training datasets and the computational costs of deploying robust models remain significant barriers. Some approaches use deep learning to leverage conditional normality for anomaly detection [3].

This paper presents a novel approach to address these challenges by combining weakly-supervised learning techniques with systematic operational practices. The work is structured around two main contributions: the development of a knowledge distillation framework for VAD [1] and the application of MLOps practices for efficient data and model management.

I Distilling Aggregated Knowledge for Weakly-Supervised VAD

VAD systems aim to identify activities that deviate from normal behaviors, as defined by the training data distribution. Traditional approaches rely heavily on frame-level annotations, which are labor-intensive and time-consuming, particularly for large-scale datasets. Recent studies have shown that weakly-supervised learning—using video-level labels—can alleviate the annotation burden. However, this technique faces challenges due to limited supervision, resulting in suboptimal anomaly detection performance.

To address this, a Temporal Aggregation Module (TAM) is proposed, which aggregates spatiotemporal knowledge from multiple pretrained backbones through self- and cross-attention mechanisms. This module integrates content and positional information, constructing a comprehensive representation of video data. Furthermore, a bi-level fine-grained knowledge distillation framework is introduced, which transfers knowledge from a computationally expensive Teacher model into an efficient Student model. By aligning prediction-level and feature-level outputs, the method enhances the performance of weakly-supervised VAD while maintaining computational efficiency.

Extensive empirical evaluation demonstrates that the proposed framework outperforms existing methods on benchmark datasets, highlighting the significance of knowledge transfer and aggregation in weakly-supervised learning scenarios.

To summarize, our primary contributions are:

1. We investigate the effectiveness of knowledge transfer from **multiple backbones** on weakly-supervised VAD and argue its importance for achieving robust performance.
2. We propose an architecture for combining and aggregating information of multiple pre-trained backbones.
3. We aim to implement a well-organized and systematic pipeline to handle versioning and deployment of our datasets and **models** for future adaptability and accessibility by incorporating MLOps.

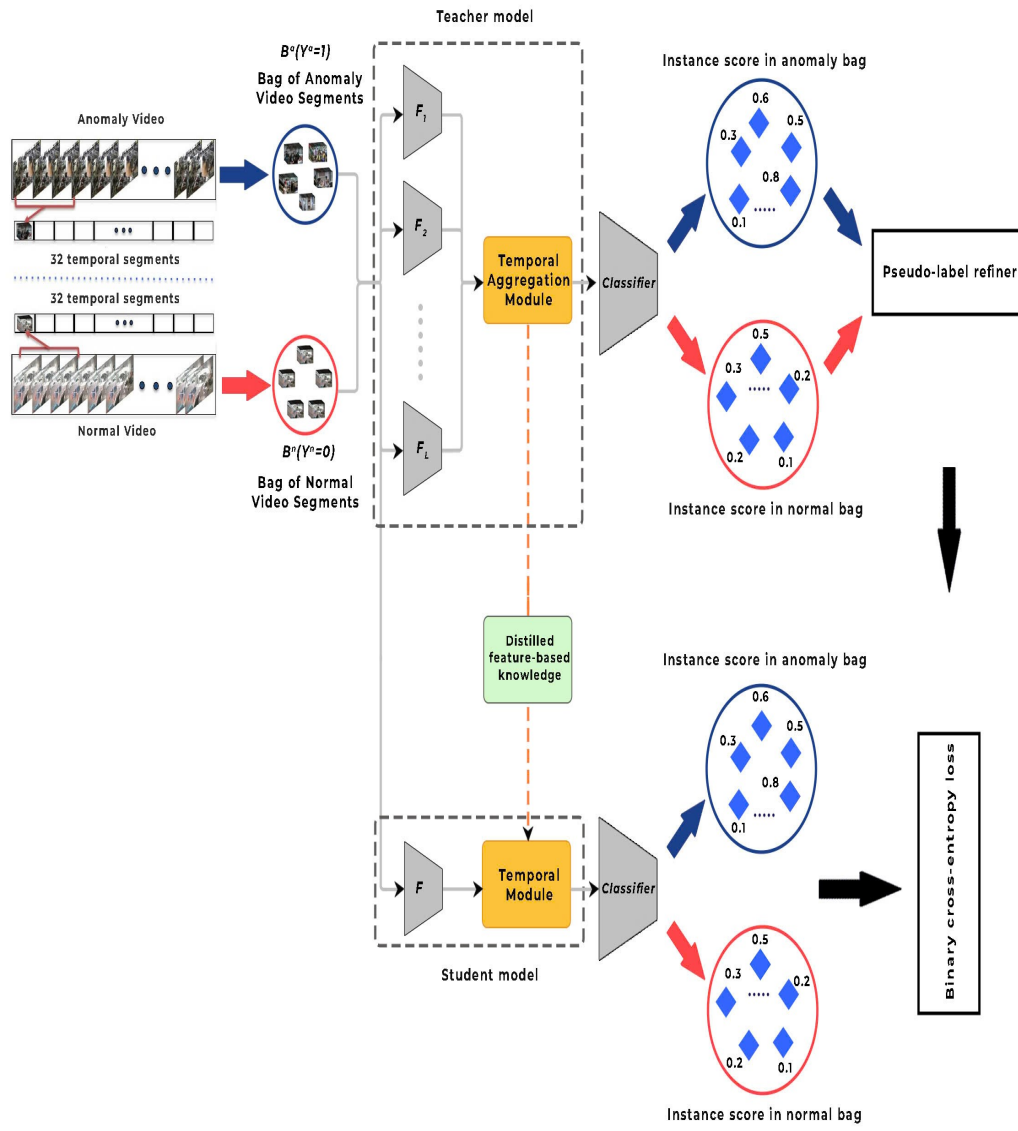


Figure 1:

B Project Management

C Related Works

I Related Works 1

1 Content 1

2 Content 2

II Related Works 2

1 Content 1

2 Content 2

III Related Works 3

1 Content 1

2 Content 2

D Materials

I Hardware

II Framework

III Tools

IV Dataset

1 Dataset 1

2 Dataset 2

3 Dataset 3

E Methods

I Method 1

II Method 2

III Method 3

IV Evaluation Metrics

In the context of Weakly-Supervised Video Anomaly Detection (VAD), the evaluation metrics are divided into two levels: **video-level metrics** (used for training/validation) and **frame-level metrics** (used for testing). These metrics measure the performance of the model in detecting anomalies either at the video level or at finer-grained frame-level granularity.

1 Video-Level Metrics

Video-level metrics are used to evaluate the model's ability to correctly classify a video as normal or anomalous during training and validation. The key metric is the **Video-Level AUC (Area Under the Curve)**.

Video-Level AUC: The AUC score measures the model's ability to distinguish between normal and anomalous videos. It is computed as the area under the Receiver Operating Characteristic (ROC) curve, which plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various thresholds.

$$\text{TPR} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (1)$$

$$\text{FPR} = \frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}} \quad (2)$$

Example: Assume the model classifies 50 anomalous videos correctly (*True Positives*), 10 anomalous videos incorrectly (*False Negatives*), 90 normal videos correctly (*True Negatives*), and 20 normal videos incorrectly (*False Positives*). The TPR and FPR are computed as:

$$\text{TPR} = \frac{50}{50 + 10} = 0.833, \quad \text{FPR} = \frac{20}{20 + 90} = 0.182$$

The AUC score integrates these values over all possible thresholds.

2 Frame-Level Metrics

Frame-level metrics are used to evaluate the model's ability to localize anomalous events within a video during testing. The primary metrics include **Frame-Level AUC** and **Average Precision (AP)**.

Frame-Level AUC: Similar to video-level AUC, the frame-level AUC measures the area under the ROC curve, but at the frame granularity. It evaluates how well the model distinguishes anomalous frames from normal frames.

Average Precision (AP): The Average Precision is calculated from the Precision-Recall (PR) curve, which plots Precision against Recall at various thresholds.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (3)$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (4)$$

The AP is computed as:

$$AP = \sum_n (R_n - R_{n-1}) P_n$$

where R_n and P_n are the recall and precision at the n -th threshold, respectively.

Example: Consider a test video where the model predicts anomaly scores for 100 frames. If 30 frames are correctly identified as anomalous (*True Positives*), 10 anomalous frames are missed (*False Negatives*), and 20 normal frames are misclassified as anomalous (*False Positives*), we compute:

$$\text{Precision} = \frac{30}{30 + 20} = 0.6, \quad \text{Recall} = \frac{30}{30 + 10} = 0.75$$

The PR curve is used to calculate the AP as the area under this curve.

3 Metric 2

4 Metric 3

F System design and implementation

I System design

II MLOps

1 Introduction

2 Semantic versioning

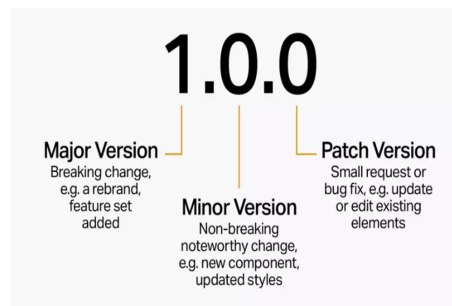


Figure 2:

SemVer Version Structure

A SemVer version is represented by three increasing integers, in the following order:

- **Major (First number)**: This is the first number in the version and is incremented whenever there is a change that is not backward compatible with the previous version. When this number is incremented, there may be a major change in the functionality or structure of the software.
- **Minor (Middle number)**: The second number represents a version that changes features or adds features without breaking backward compatibility.
- **Patch (Last number)**: The third number increases when there are patches or minor modifications that do not change functionality or are backward compatible.

A SemVer version has the following structure: Major.Minor.Patch

So in summary:

- **MAJOR**: Represents a major change, incompatible with the old version.
- **MINOR**: Represents minor changes, still compatible with the old version. Different from PATCH, instead of updating something, MINOR can create something.
- **PATCH**: Represents minor changes, still compatible with the old version, usually related to updates or minor bug fixes. This reminds me of PUT and PATCH in HTTP request methods.



Figure 3: Example of Semantic Versioning.

- **X** stands for Major version: If the current version is 2.6.8 then the next major upgrade to the version will be 3.0.0. Increase the value of X when breaking existing API.
- **Y** stands for Minor version: If the current version is 2.6.8 then the next minor upgrade will be 2.7.0. Increase the value of Y when implementing new features in a backward compatible manner.

- **Z** stands for Patches version: If the current version is 2.6.8 then the next version to upgrade the patch to will be 2.6.9. There is no limit to these numbers. Increase the value of Z when fixing bugs.

Hyphens and special characters

SemVer also allows the use of hyphens and special characters to describe additional information:

- **Dash:** Use a dash (-) to add additional information after the version number. For example: 1.0.0-alpha, 2.0.0-beta.3.
- **Special characters:** Use the special character (+) to add additional information, usually related to modifications in the source code. For example: 1.0.0+build123.

Meaning and Advantages

Semantic Versioning has several important advantages:

- **Easy to read and understand:** SemVer allows users and developers to easily understand the features and changes in a software version just from the version number.
- **Compatibility:** Versions with increasing Minor or Patch numbers only propose changes that do not break backward compatibility, allowing users to upgrade software without worrying about problems.
- **Easy version management:** SemVer helps developers manage and track software versions efficiently.
- **Auto support:** Many automated package management tools and systems support SemVer, making version management more automated and reliable.

SemVer has become an important standard in the software development industry and is a useful tool for version management and enhancing compatibility in open source and commercial software development.

Points to note

- The first version starts at 0.1.0 and not 0.0.1 because no bug fixes have been made yet, instead we start with a feature set as the first draft of the project.
- Before 1.0.0 it was just the Development Phase where you focused on getting things done. This phase was for developers where the system was being developed.

SemVer does not include libraries tagged 0.*.*. The first stable version is 1.0.0.

How to apply Semantic Versioning

Semantic Versioning has the following structure:

MAJOR.MINOR.PATCH

Trong đó:

- **MAJOR:** Major change, not compatible with old version.
- **MINOR:** Add new features while remaining compatible.
- **PATCH:** Minor bug fixes or improvements that do not affect core functionality.

Examples applied to assignments

- **MAJOR:**

- Convert from S3D to another model (e.g., I3D or R3D).
- Use the new dataset instead of the old one.

- **MINOR:**

- Add new loss function or improve anomaly prediction results.
- Retrain S3D with a larger dataset.

- **PATCH:**

- Parameter optimization (e.g., learning rate, batch size).
- Fix minor bugs in feature extraction or in training pipeline.

3 Data management

content...

Semantic Versioning for Datasets

Semantic Versioning (MAJOR.MINOR.PATCH) is applied to manage the versioning of datasets as follows:

MAJOR (Significant Changes)

- Changes in the data format (e.g., from JSON to CSV).
- Changes in the large data structure (e.g., renaming or deleting data fields).
- Addition or removal of a large portion of data.
- Modifications in the labeling criteria.

MINOR (Improvements)

- Adding new data samples while maintaining the current structure.
- Adding additional descriptive information or new labels.
- Creating new data subsets (e.g., adding a test set).

PATCH (Bug Fixes)

- Fixing errors in mislabeled data.
- Handling missing values or corrupted files.
- Updating metadata without changing the main data.

Examples

- **1.0.0**: The first version of the dataset.
- **1.1.0**: Adding new data samples or additional information without changing the structure.
- **1.1.1**: Fixing minor issues such as incorrect labels or missing values.

Component	Meaning	When to Increase Version	Example
Major	Major change, incompatible with previous versions.	Change in data format, switching to a different dataset.	1.0.0 → 2.0.0
Minor with previous versions.	Adds new features or data, still compatible Adding metadata, adding PLR labels, or improving labels.	1.0.0 → 1.1.0	
Patch	Small bug fixes, performance optimizations, or data updates that do not change the main structure.	Fixing incorrect labels, updating small data files, no structural changes.	1.0.0 → 1.0.1

Table 1: Semantic Versioning for Data

4 Model management

Semantic Versioning for Model

Semantic Versioning (MAJOR.MINOR.PATCH) is applied to manage the versioning of machine learning models as follows:

MAJOR (Significant Changes)

- Changes in the model architecture (e.g., from CNN to Transformer).
- Modifications in input/output formats or tensor dimensions.
- Transition to a new dataset or a different training pipeline.
- Dropping support for some old features.

MINOR (Improvements)

- Adding new mechanisms (e.g., introducing an Attention Mechanism).
- Enhancing accuracy, performance, or improving the training process.
- Providing pre-trained weights for specific applications.

PATCH (Bug Fixes)

- Fixing bugs in the model architecture.
- Fine-tuning hyperparameters.
- Minor optimizations that do not affect the main functionality.

Examples

- **2.0.0**: The second major version with a complete architecture overhaul.
- **2.1.0**: Adding new features or improving performance.
- **2.1.1**: Fixing minor bugs in the previous version.

Component	Meaning	When to Increase Version	Example
Major	Major change, incompatible with previous versions.	Change in model architecture, introducing a new model type.	1.0.0 → 2.0.0
Minor	Adds new features or model improvements, still compatible with previous versions.	Improvement in accuracy, adding new functionalities.	1.0.0 → 1.1.0
Patch	Bug fixes, performance improvements, or small tweaks that do not impact the overall model.	Fixing small bugs in training, enhancing model's inference speed.	1.0.0 → 1.0.1

Table 2: Semantic Versioning for Model

5 Model Serving

G Results

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H Discussion

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I Conclusion

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J Testing Page

Test glossary: optimizer

Test acronym: Gradient Descent (GD)

Test citation:

a/ [einstein]

b/ [dirac]

Acronym

G

GD: Gradient Descent.

Glossary

O

optimizer: Some desc for optimizer.

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